AN ENGINEERING PROJECT PROPOSAL

On

DiGiWallet:
Digital Academic Credential Verification System

Submitted By

Bibek Ghimire - 230309

Maheshwar Pant - 230322

Manisha Oli - 230347

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0.1 Abstract

DiGiWallet is a software-based academic credential verification system designed to reduce repeated document submission, manual verification delays, and the risk of fake or tampered academic certificates in Nepal. The system enables academic institutions to issue digitally signed credentials that students can store in a digital wallet and share through a QR code or verification link. The credential data is converted into a structured digital format, hashed using cryptographic techniques, and signed with the issuer's private key. Verifiers such as employers or institutions can instantly check the authenticity of the credential using the issuer's public key and identify whether it is valid, invalid, tampered, revoked, or not found [1,2]. The project will be developed using the MERN stack, with React.js for the frontend, Node.js and Express.js for backend services, and MongoDB for data storage. The expected outcome is a functional prototype that demonstrates secure, reusable, and QR-based academic credential issuance and verification in a Nepal-focused context.

0.2 List of Abbreviations

- **API**: Application Programming Interface
- **CSS**: Cascading Style Sheets
- **DG**: Digital Graduate / Digital Credential Wallet
- **HTML**: HyperText Markup Language
- **JWT**: JSON Web Token
- **KYC**: Know Your Customer
- **MERN**: MongoDB, Express.js, React.js, Node.js
- **PKI**: Public Key Infrastructure
- **QR**: Quick Response
- **SHA**: Secure Hash Algorithm
- **UI**: User Interface
- **VC**: Verifiable Credential
- **SSI**:Self-Sovereign Identity
- **MIT**:Massachusetts Institute of Technology
- **DID**:Decentralized Identifiers

0.3 List of Figures

Contents

Chapter 1

Introduction

1.1 Background Introduction

Academic credentials such as certificates, transcripts, mark sheets, and degree documents are important proof of a student's educational achievement. In Nepal, students are often asked to submit photocopies or scanned copies of these documents for jobs, internships, higher studies, scholarships, professional registration, and other official purposes. This makes the process repetitive and creates unnecessary dependency on manual document verification.

Employers and institutions also need to confirm whether the academic documents submitted are genuine. In many cases, they have to contact the concerned college or university through email, phone, or official communication. For example, Tribhuvan University still provides certificate verification through email contact, which shows that manual or semi-digital verification is still practiced in Nepal. This process can be slow and uncomfortable, especially when quick verification is required [3].

At the same time, Nepal is slowly moving towards digital transformation.The Digital Nepal Framework has identified education as one of the important sectors for digital development and aims to improve services through technology [4].Based on this need, **DiGiWallet** is proposed as a secure and reusable academic credential verification system. Instead of submitting photocopies again and again,students can receive a digitally signed credential once and share it whenever needed through a QR code or verification link.

1.2 Statement of Problem

Academic credential verification in Nepal is still slow, repetitive, and highly dependent on manual verification. Students often submit the same certificates multiple times, while employers and institutions face difficulties confirming whether those documents are genuine.

- Students need to provide photocopies or scanned copies of academic documents for jobs, internships, higher studies, KYC, and other verification processes.

- Employers often have to contact colleges or universities manually, which makes verification time-consuming and inconvenient.
- Physical or scanned certificates can be misused, altered, or faked, creating trust issues and showing the need for a secure digital verification system like **DiGiWallet**.

1.3 Objectives

The objectives of DiGiWallet Project are:

- To develop a secure web-based system for issuing digitally signed academic credentials.
- To provide students with a digital wallet for storing and sharing credentials through QR codes or verification links.
- To enable employers and institutions to verify credentials instantly.
- To reduce repeated document submission, manual verification delays, and fake certificate risks.
- To demonstrate a reliable prototype using web and cryptographic technologies.

1.4 Scope of the Project

The scope of DiGiWallet is focused on developing a web-based prototype for issuing, storing, sharing, and verifying digital academic credentials. The system will mainly include four modules: Admin Panel, College/Issuer Portal, Student Wallet, and Verifier Portal.

The Admin Panel will be used to manage colleges, users, public keys, and system records. The College/Issuer Portal will allow authorized colleges to enter verified student academic information and issue digitally signed credentials. The Student Wallet will help students view their issued credentials and share them easily through a QR code or verification link. Similarly, the Verifier Portal will allow employers or institutions to check the authenticity of a credential instantly.

In the initial version, the project will be developed as a working prototype and will not be directly connected with real government systems, national identity systems, or actual university databases. Instead, sample or controlled academic data will be used for testing and demonstration. Advanced features such as blockchain integration, decentralized identity, biometric verification, and national-level implementation will be considered as future enhancements[5].

Chapter 2

Literature Review

2.1 Introduction

Academic fraud has become a significant concern worldwide, as fake degrees and manipulated certificates can negatively affect educational institutions, employers, and students.The rapid growth of digital technology in the education sector has transformed the way academic records are created, stored, and verified.The increasing use of digital technologies in education system has created a need for secure and efficient academic credential verification systems. Traditional paper-based certificates are vulnerable to forgery, damage, and loss. Manual verification methods are also time-consuming and costly for universities and employers. To address these issues, researchers have proposed Digital Academic Credentials Verification Systems.A Digital Academic Credentials Verification System is designed to issue, manage, and verify academic certificates electronically using secure technologies such as digital signatures, QR codes, databases, and blockchain.

2.2 Related Work and Theory

In the last ten to fifteen years,the production of forged academic certificates has become a global problem as educational qualifications have gained increasing commercial value[6]. The rising number of fake diplomas is often seen as a consequences of the crisis, where desperate people forge certificates in order to obtain job qualifications[7]. To solve this problem and to improve the burdensome and time-consuming administrative work about certificates, many researchers have proposed blockchain technology to speed-up academic diploma issuance and verification[8,9,10,11]. Traditional academic verification systems rely heavily on manual processes involving physical documents, institutional databases, and third-party verification agencies. These approaches are often slow, expensive, and prone to human error. Caldarelli and Ellul (2021)observed that centralized credential systems suffer from limited transparency, data manipulation risks, and inefficient verification procedures, especially in cross-border education and employment context.

As the ”Great Resignation” continues in Europe, the demand for fast and trustworthy hiring processes has become increasingly severe [12]. Kumar et al. [13] developed a system that uses blockchain to store and verify educational credentials, with the aim of reducing the time required for employers to authenticate these credentials. Similarly, Truver [14] uses Ethereum to create an immutable record of academic achievements. BCERT [15] is a decentralized academic certificate system that uses blockchain to improve credential security and verification efficiency. According to Caldarelli and Ellul (2021), centralized credential systems face several challenges including lack of transparency, inefficient verification procedures, and high administrative overhead[16]. Rustemi et al. (2023) conducted a systematic literature review on blockchain-based academic certificate verification systems and found that blockchain improves transparency, security,traceability, and trust[17]. The study explained that storing certificate hashes on blockchain helps prevent tampering and unauthorized modification. Tariq et al. (2019) proposed a blockchain-based degree verification framework called Cerberus. Their system used smart contracts for automated verification and credential revocation[18]. The research showed that blockchain can reduce verification time and eliminate dependency on third-party verification agencies. Herbke et al. (2024) developed a decentralized credential management framework using DIDs and verifiable credentials[19]. Their system allowed students to securely share academic achievements without relying on centralized verification authorities.

Recent research has focused on VCs and DIDs for academic credential management. These technologies are based on the concept of Self-Sovereign Identity (SSI), where users have full control over their digital credentials. Platforms such as Accredible, Blockcerts, and Credly provide online credential issuance and verification services.Blockcerts[20], developed by MIT Media Lab and Learning Machine, is an open standard for blockchain-based certificates. The platform stores certificate hashes on blockchain networks and allows independent verification without contacting the issuing institution.

2.3 Challenges and Research Gap

The digital credential verification system has evolved significantly, but still it faces some challenges like scalability issues, high implementation cost, lack of standardization and technical complexity. There is significant progress in blockchain-based academic credential verification system; however several research gaps remain. Many proposed system focuses primarily on security and immutability while providing limited support for interoperability and large scale deployment. In addition to it,most studies remain conceptual or prototype-based, with limited real-world implementation and evaluation. There is also limited research on integrating Verifiable Credentials, decentralized identity systems, QR-based verification, and scalable blockchain infrastructures into a unified academic credential ecosystem suitable for developing countries and multi-institution environments. So, further research is needed to overcome these gaps.

Chapter 3

Project Methodology

The proposed DiGiWallet system will be developed as a web-based prototype for issuing, storing, sharing, and verifying academic credentials digitally. The methodology focuses on the technical process required to build the system, including system architecture, module development, cryptographic credential verification, QR-based sharing, tools required, and testing. Since this is a proposal for a minor project, the first version will focus on developing the core working prototype, while later versions can improve security, scalability, user experience, and institutional integration.

3.1 System Development Method

The system will be developed using an Agile development approach with iterative and incremental implementation. This method is suitable for DiGiWallet because the project can be divided into small functional modules and improved step by step [**?**].

In the first iteration, the core prototype will be developed for the minor project. This version will include user authentication, college credential issuance, student wallet, QR code generation, and credential verification. After testing the first version, further improvements such as revocation management, verification logs, stronger security, better dashboard design, and future integration features can be added in the next version.

The development process will follow the following stages:

1. **Requirement Finalization:** Identify the main users, modules, and minimum features required for the prototype.
2. **System Design:** Design the architecture, database structure, user roles, and credential verification logic.
3. **Module Development:** Develop the admin panel, college issuer portal, student wallet, and verifier portal incrementally.
4. **Integration:** Connect frontend, backend, database, authentication, QR generation, and cryptographic verification.

5. **Testing and Improvement:** Test the system with sample academic data and improve the system based on errors, feedback, and project requirements.

3.2 Technical Architecture Method

The technical architecture of DiGiWallet will follow a client-server model using the MERN stack. The frontend will provide separate interfaces for admin, college, student, and verifier. The backend will handle authentication, credential issuance, digital signature generation, QR code creation, verification logic, and communication with the database.

The system architecture will be divided into the following layers:

- **Presentation Layer:** This layer will contain the user interfaces for admin, college, student, and verifier. It will be developed using React.js.
- **Application Layer:** This layer will contain the main backend logic such as login, role-based access, credential issuance, QR generation, signature verification, and revocation handling. It will be developed using Node.js and Express.js.
- **Database Layer:** This layer will store users, colleges, credentials, public keys, QR links, credential status, and verification logs using MongoDB.
- **Security and Verification Layer:** This layer will handle password hashing, JWT authentication, credential hashing, digital signature creation, and public-key-based verification [**?**, **?**, **?**].

This architecture will make the system modular, understandable, and suitable for future improvement.

3.3 System Modules

The proposed system will be developed through the following major modules:

- **Admin Module:** Manages colleges, users, public keys, and system-level records.
- **College / Issuer Module:** Allows authorized colleges to enter verified student academic data and issue digitally signed credentials.

- **Student Wallet Module:** Allows students to view their issued credentials and share them using QR codes or verification links.
- **Verifier Module:** Allows employers or institutions to verify credentials by scanning a QR code or opening a verification link.
- **Credential Verification Module:** Checks the digital signature, credential hash, issuer public key, and credential status.
- **Revocation Module:** Allows a previously issued credential to be marked as revoked when incorrect data, duplicate records, or administrative errors are found.

3.4 Technical Workflow

The technical workflow of DiGiWallet will explain how an academic credential moves from the college issuer to the student wallet and finally to the verifier.

The proposed workflow is as follows:

1. Admin registers and authorizes the college in the system.
2. College logs in to the issuer portal.
3. College enters verified student academic information.
4. System converts the data into a structured digital credential format.
5. System generates a hash of the credential data using a secure hash algorithm such as SHA-256 [**?**].
6. The hash is signed using the college private key according to the digital signature concept [**?**].
7. The signed credential is stored in the database.
8. The credential is assigned to the student wallet.
9. System generates a QR code or verification link.
10. Student shares the QR code or verification link with a verifier.
11. Verifier scans the QR code or opens the link.
12. System checks the credential using the issuer public key.
13. System displays the result as valid, invalid, tampered, revoked, or not found.

3.4.1 Technical Workflow Diagram

The following section is reserved for the workflow diagram of the proposed DiGiWallet system. The diagram will visually represent the process of credential issuance, student sharing, and verifier-side validation.

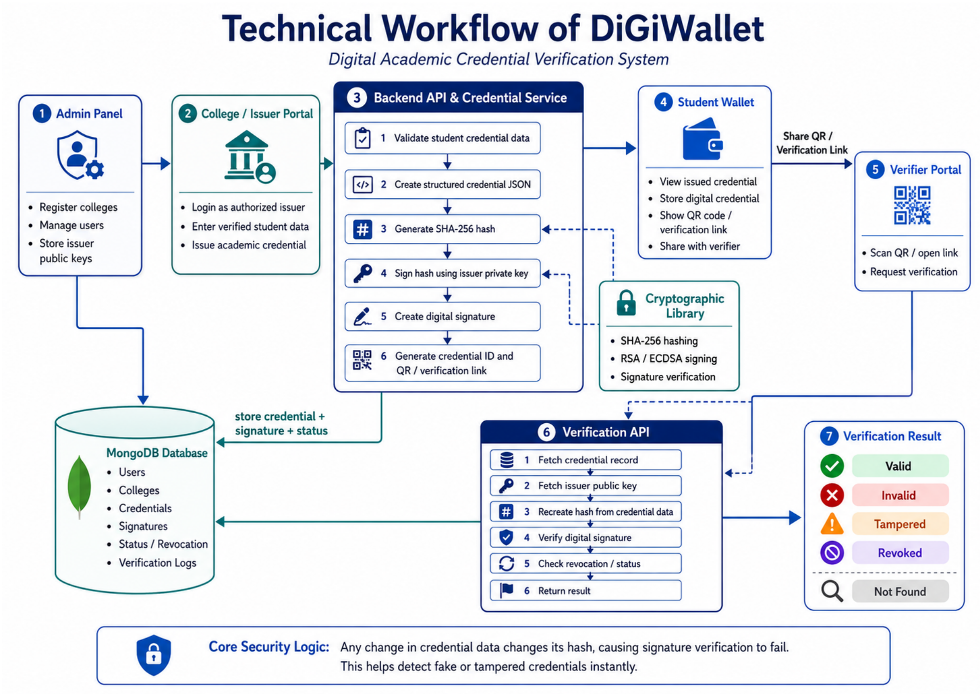

Figure 3.1: Technical Workflow of DiGiWallet System

3.5 Tools Required

The following tools and technologies will be required to develop the proposed system:

- **Frontend:** React.js, HTML, CSS, Tailwind CSS
- **Backend:** Node.js and Express.js
- **Database:** MongoDB
- **Authentication:** JWT and bcrypt
- **Cryptography:** Node.js Crypto module or suitable cryptographic library
- **QR Code:** QR code generation library
- **API Testing:** Postman

- **Code Editor:** Visual Studio Code
- **Version Control:** Git and GitHub
- **Deployment:** Initially deploy locally for development and testing, and later deploy on Vercel

3.6 Testing and Validation Method

The system will be tested using sample academic credential data. Testing will focus on checking whether each module performs its expected function correctly.

The following cases will be tested:

- User login and role-based access control.
- College credential issuance.
- Credential hash generation.
- Digital signature creation.
- QR code and verification link generation.
- Student wallet credential display.
- Verifier-side credential verification.
- Detection of tampered credential data.
- Revoked credential status checking.
- Invalid or not-found credential handling.

The final prototype will be considered successful if a college can issue a digitally signed credential, a student can share it using a QR code or verification link, and a verifier can check its authenticity through the system.

Chapter 4

Expected Outcomes

4.1 Expected Outcome

At the end of the project, the following outputs are expected:

- A working web-based prototype of DiGiWallet.
- Admin dashboard for managing colleges, users, public keys, and credentials.
- College/Issuer Portal for issuing digitally signed academic credentials.
- Student Wallet for viewing and sharing issued credentials.
- QR-based verification link for each credential.
- Verifier Portal for instant credential verification.
- Detection of valid, invalid, tampered, revoked, and not-found credentials.
- Database design for users, colleges, credentials, signatures, and verification logs.
- Documentation of system architecture, workflow, security model, and future scope.

The final system will demonstrate how a digitally signed academic credential can reduce repeated document submission and make verification faster, more reliable, and more secure.

Chapter 5

Timeline/Work Schedule

Work Schedule Table

Activities	week1	week2	week3	week4	week5	week6	week7	week8	week9	week10	week11	week12
Project title selection and idea finalization												
Requirement analysis and literature review												
System architecture and database design												
Authentication and role-based access control												
Admin and college issuer portal development												
Hashing and digital signature implementation												
Student wallet and QR code generation												
Verifier portal and signature verification												
Revocation and verification log module												
Testing and security validation												
UI polishing and deployment												
Documentation and final presentation												

Chapter 6

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